# GRADE 11 TECHPRO ACTIVITY SHEET

## RESPONSIVE IMAGE GALLERY USING FLEXBOX

**ICT / COMPUTER PROGRAMMING | GRADE 11 | VISUAL STUDIO CODE**

Name: _____ Section: _____ Date: _____

### LEARNING OBJECTIVES

Create an image gallery using semantic HTML elements.

Use display: flex, flex-wrap, justify-content, and gap to arrange gallery cards.

Style images using width, height, and object-fit.

Use a media query to adapt the gallery to smaller screens.

Provide meaningful alternative text (alt text) for images.

### WHAT YOU NEED

Computer with Visual Studio Code and a web browser.

Create a folder named Responsive_Image_Gallery and a file named index.html.

Copy the starter code below, save it, and open index.html in a browser.

The sample image links are placeholders and need an internet connection. Replace them with your own image files if preferred.

### CSS CONCEPTS

display: flex turns an element into a flex container; its direct children become flex items.

flex-wrap: wrap allows cards to move onto another row when space is limited.

justify-content: center centers cards along the main axis. gap creates consistent spacing.

flex: 1 1 220px lets a card grow, shrink, and start with a 220-pixel basis.

max-width prevents a card from becoming too wide.

width: 100% makes an image fill its card width. object-fit: cover fills the image area while preserving proportions; edges may be cropped.

@media (max-width: 600px) applies rules when the viewport is 600 pixels wide or narrower.

### STARTER CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>TechPro Image Gallery</title>
<style>
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #f5f5f5; }
header { background: #7b1530; color: white; text-align: center; padding: 24px; }
.gallery { display: flex; flex-wrap: wrap; justify-content: center; gap: 16px; max-width: 1100px; margin: auto; padding: 20px; }
.gallery-item { flex: 1 1 220px; max-width: 320px; background: white; border: 2px solid #7b1530; border-radius: 8px; overflow: hidden; }
.gallery-item img { width: 100%; height: 200px; object-fit: cover; display: block; }
.caption { padding: 12px; text-align: center; }
footer { background: #222; color: white; text-align: center; padding: 14px; }
@media (max-width: 600px) {
  .gallery { flex-direction: column; align-items: center; }
  .gallery-item { width: 100%; max-width: 420px; }
}
</style>
</head>
<body>
<header><h1>TechPro Image Gallery</h1><p>Responsive Image Gallery using Flexbox</p></header>
<main>
<section class="gallery">
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Computer+Lab" alt="Computer laboratory">
<div class="caption">Computer Laboratory</div>
</article>
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Programming" alt="Programming activity">
<div class="caption">Programming Activity</div>
</article>
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Robotics" alt="Robotics project">
<div class="caption">Robotics Project</div>
</article>
```

```
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Networking" alt="Networking lesson">
<div class="caption">Networking Lesson</div>
</article>
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=Web+Design" alt="Web design activity">
<div class="caption">Web Design</div>
</article>
<article class="gallery-item">
<img src="https://placehold.co/600x400?text=ICT+Projects" alt="ICT student projects">
<div class="caption">ICT Student Projects</div>
</article>
</section>
</main>
<footer>Grade 11 TechPro | Responsive Image Gallery</footer>
</body>
</html>
```

## ACTIVITY 1 — BUILD AND VIEW

1. Open the folder in Visual Studio Code and open index.html.
2. Study the HTML structure: header, main, section, article, image, caption, and footer.
3. Save the file and open it in a browser.
4. Identify the gallery container and the six gallery items.
5. Check that every image has an alt attribute describing its content.

## ACTIVITY 2 — EXPERIMENT WITH FLEXBOX

1. Find .gallery in the CSS.
2. Test justify-content values: flex-start, center, flex-end, space-between, space-around, and space-evenly.
3. Restore center. Change gap from 16px to 30px, compare, then restore 16px.
4. Change flex-wrap: wrap to nowrap. Narrow the browser and observe what happens. Restore wrap.
5. Record: Why is flex-wrap: wrap useful for an image gallery?

## ACTIVITY 3 — CONTROL CARD SIZE

1. Find flex: 1 1 220px in .gallery-item.
2. Change 220px to 280px. Observe how many cards fit in each row.
3. Change max-width: 320px to max-width: 260px. Compare card widths.
4. Restore the original values.
5. Record: What is the effect of changing the flex-basis value?

## ACTIVITY 4 — STYLE THE IMAGES

1. Find .gallery-item img.
2. Change height: 200px to height: 250px.
3. Test object-fit: contain, fill, and cover. Compare the results.
4. Restore height: 200px and object-fit: cover.
5. Record: Which object-fit value fills the image area while maintaining proportions?

## ACTIVITY 5 — TEST RESPONSIVE DESIGN

1. Find the @media (max-width: 600px) rule.
2. Change 600px to 700px and resize the browser around that width.
3. Temporarily remove flex-direction: column from .gallery inside the media query. Observe the difference.
4. Restore flex-direction: column and align-items: center.
5. Test desktop, tablet-like, and narrow mobile widths.
6. Record: What is a breakpoint, and why should a gallery be tested at different widths?

## MINI-PROJECT — MY ICT PHOTO GALLERY

Create a gallery with at least twenty images related to yourself, school, ICT, programming, technology, or student projects.

☐ Use a section with class gallery and article elements with class gallery-item.

☐ Use Flexbox with flex-wrap: wrap and a consistent gap.

☐ Give every image meaningful alt text and a caption.

☐ Use object-fit: cover and a consistent image height.

☐ Add a media query for small screens.

☐ Use a maroon, white, and light-gray color theme.

☐ Add a page title and footer.

☐ Check that cards do not overlap and there is no unwanted horizontal scrolling.

## CHECKLIST AND REVIEW

☐ My index.html opens correctly in a browser.

☐ My gallery contains at least eight images.

☐ Cards wrap into additional rows when needed.

☐ Images have consistent display dimensions and meaningful alt text.

☐ The gallery adapts to narrow screens.

1. What is the role of display: flex in the gallery?

   _____

2. What does flex-wrap: wrap do?

   _____

3. What is the purpose of gap?

   _____

4. What is the difference between width: 100% and object-fit: cover?

   _____

5. Why should images have meaningful alt text?

   _____

6. What does a media query do in this project?

   _____

## SUBMISSION AND REFERENCE

Suggested filename: Grade11_Name_Responsive_Image_Gallery.html

W3Schools CSS Flexbox: https://www.w3schools.com/css/css3_flexbox.asp

W3Schools Flexbox Container: https://www.w3schools.com/css/css3_flexbox_container.asp

W3Schools Responsive Flexbox: https://www.w3schools.com/css/css3_flexbox_responsive.asp